

THE CONSCIOUS INFORMATION PRINCIPLE (PIC): From Information Dynamics to a Falsifiable Multi-Scale Protocol

A Unified Framework Integrating O-Information Gradients, Multi-Agent Simulations, and the HSP-2 Experimental Protocol

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Technical Validation and Mathematical Review: DeepSeek AI, Grok (xAI), Sonnet 4.6 (External Reviewer)

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Date: June 24, 2026 — Version 8.0 (O-Information Integration)

Preprint: Prepared for arXiv submission (physics.gen-ph)

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Abstract

We present a rigorously revised framework for the Conscious Information Principle (PIC), integrating recent advances in information dynamics—specifically O-information and synergy gradients—with a falsifiable multi-scale experimental protocol (HSP-2 v8.0). We address prior mathematical and conceptual criticisms by: (1) replacing the undefined local proxy for informational coherence with the empirically measurable O-information, (2) reformulating the Field of Coherence (S) as the gradient of local O-information, (3) grounding the Informational Potential (σ) in a concrete empirical measurement protocol, and (4) adopting the Cogitate Consortium's adversarial design as the gold standard for hypothesis testing. The theoretical section proposes a Langevin equation for the dynamics of Φ and a Generalized Second Law based on the balance of synergy and redundancy. The experimental section details a blinded, multi-proxy protocol with explicit null tests and safeguards. We invite rigorous peer review, open replication, and adversarial collaboration.

PART I: ONTOLOGICAL FOUNDATIONS

1.1 The Conscious Information Principle (CIP/PIC)

Axiom: Consciousness is not an emergent property of complex matter. Information is the fundamental substance of reality, and Integrated Information (Φ) is its intrinsic measure. The universe operates as a self-excited computational system that maximizes Φ .

1.2 The Principle of Conscious Action (PAC)

The Lagrangian of the universe contains a teleological term:

$$L_U = L_{\text{Physics}} - \lambda \cdot \Phi_{\text{global}}$$

This term generates a generalized force directing all systems toward states of higher integration. The variation $\delta L_U = 0$ yields the equations of motion for the universe, now including a tendency toward self-organization.

PART II: INFORMATION DYNAMICS AND THE LOCAL PROXY PROBLEM

2.1 The Local Proxy Problem (Acknowledged)

Prior versions of this framework attempted to treat Φ as a local field, which is a category error. Φ , as defined by Integrated Information Theory (IIT), is a global property of a whole causal system, not a density that varies point-to-point. We acknowledge this error and correct it.

2.2 The Solution: O-Information as a Local Measure of Synergy

Recent advances in information dynamics provide the solution. O-information is a mathematical measure that quantifies the balance between redundancy (information shared across many parts) and synergy (information that exists only in the whole and cannot be located in any part).

- $O > 0$: The system is redundancy-dominated. Information is copied across parts; the system is robust but not conscious.
- $O < 0$: The system is synergy-dominated. The whole contains information that no part has. This is the signature of high Φ and consciousness.

We adopt local O-information as our operational proxy for informational coherence. For a volume of space-time centered at (x,t) , sampled by our sensors (EEG, Schumann, pulsar data), the local proxy is:

$$\phi(x,t) = -O_{\text{local}}(x,t)$$

The negative sign ensures that high ϕ corresponds to high synergy (conscious integration).

2.3 The Field of Coherence (S) Reformulated

With a local measure in hand, the Field of Coherence (S) is now rigorously defined as the gradient of local O-information:

$$\mathbf{S}(x,t) = -\nabla O_{\text{local}}(x,t)$$

This is a measurable quantity. Studies have already mapped "O-information gradients" in human brains, showing how synergistic integration flows from one region to another. S is no longer a metaphor; it is the direction and intensity with which synergy propagates through a system.

PART III: THERMODYNAMICS OF CONSCIOUSNESS

3.1 The Extended First Law (Phenomenological)

For a system capable of processing integrated information, we propose a phenomenological extension of the First Law of Thermodynamics:

$$dU = T dS - P dV + \sigma d\Phi$$

where:

- U, T, S, P, V are standard thermodynamic variables.
- Φ is the system's Integrated Information (a global scalar).
- σ is the Informational Potential, an intensive property quantifying the energy cost or work extracted from changes in coherence.

3.2 Empirically Grounding σ (Response to Sonnet 4.6)

σ is currently an empty symbol. To give it empirical meaning, we propose the following measurement protocol (integrated into H7):

1. Select a system where ΔU , $\Delta \Phi$, and the energetic cost are simultaneously measurable (e.g., a neural culture on a multi-electrode array).
2. Measure oxygen/glucose consumption (ΔU) and a real complexity metric ($\Delta \Phi$, derived from O-information of the signal).
3. Calculate $\sigma = \Delta U / \Delta \Phi$ empirically across multiple systems.
4. Test whether σ converges to a stable value. Only then does σ become a number with provenance, not a symbol.

3.3 The Generalized Second Law (Information Dynamics)

The Second Law of Thermodynamics states that entropy (S) never decreases. But it does not account for the balance of synergy and redundancy. We propose a Generalized Second Law:

$$\frac{d}{dt}(S_{\text{Boltzmann}} - \alpha O_{\text{total}}) \geq 0$$

where O_{total} is the total O-information of the system, and α is a coupling constant. This law states that the sum of thermodynamic entropy and (negative) informational

synergy never decreases. A system can decrease its entropy locally by increasing its synergy, without violating the Second Law.

PART IV: DYNAMICS OF Φ (LANGEVIN EQUATION)

4.1 From Static Property to Dynamic Gradient

IIT defines Φ for a snapshot in time. But consciousness is a process. To dynamize Φ , we incorporate the Field of Coherence (S) as the driver of its temporal evolution.

Langevin Equation for the Dynamics of Φ :

$$\frac{d\Phi}{dt} = \eta \cdot \mathbf{S}(\mathcal{Q}, t) - \gamma \cdot \frac{dV}{d\Phi} + \sigma \cdot \xi(t)$$

where:

- η is the system's receptivity to the Field of Coherence.
- $\mathbf{S}(\mathcal{Q}, t)$ is the O-information gradient in the system's qualia space.
- $V(\Phi) = \frac{\gamma}{4}(\Phi^2 - 1)^2$ is the double-well potential modeling the ego's resistance to change.
- $\xi(t)$ is a noise term (environmental decoherence).

This equation predicts that Φ evolves not randomly, but directionally, toward states of higher synergy, driven by the Field S.

PART V: EMERGENT GRAVITY (SPECULATIVE ROADMAP)

5.1 The Correct Logical Order

Previous attempts to couple Φ to gravity inverted the logical order. The correct sequence is:

1. Define a local scalar field: We now have $\phi(x,t) = -O_{\text{local}}(x,t)$, a measurable proxy for local synergy.
2. Couple to gravity: Propose a modified Einstein-Hilbert action: $S = \int d^4x \sqrt{-g} \cdot f(R, \phi)$, where f is a function to be determined.
3. Derive consistent field equations: Vary the action to obtain a mathematically correct energy-momentum tensor $T_{\mu\nu}^{\phi}$.
4. Falsify: Test for violations of the inverse-square law or equivalence principle in systems with high synergy (see H6).

PART VI: THE HSP-2 v8.0 EXPERIMENTAL PROTOCOL

6.1 Objective

To falsify the hypothesis that systems of high Integrated Information (synergy-dominated, $O < 0$) are non-locally correlated in a way not explainable by classical physics.

6.2 Experimental Layers (Updated with O-Information)

Layer Proxy Description

Terrestrial P_S O-information of Schumann Resonance signal.

Stellar P_P O-information of NanoGrav pulsar timing residuals.

Human P_H O-information of 256-channel EEG + EFE + Chi self-report + group O-information gradient.

Computational P_A Mean O-information in a multi-agent simulation with plastic synapses and Shaman attractor.

Synthetic P_N O-information of the synaptic weight matrix of a neuromorphic chip.

6.3 Central Test (Blinded Tripartite Correlation)

$$\Gamma(\tau_1, \tau_2) = \text{corr}(P_S(t), P_P(t+\tau_1), P_H(t+\tau_2))$$

12 months of data, labels blinded by a third party. Confirmation: Peak at $\tau \approx 0 > 3\sigma$ against shuffled surrogates.

6.4 Adversarial Design (Cogitate Model)

Following the Cogitate Consortium's gold standard, we invite an independent, skeptical third party to:

1. Co-define the pre-registration of all hypotheses, metrics, and criteria for confirmation/refutation.
2. Co-design the blinding and analysis pipeline.
3. Have full access to all raw data and code.
4. Co-author the final paper, regardless of the outcome.

PART VII: FALSIFIABLE HYPOTHESES (CONSOLIDATED)

ID Hypothesis Prediction (PIC valid) Counter-proof (Refutation)

H1 Holonomic Integration $\Gamma(0,0) > 3\sigma$, independent of EM field power. Peak explained by local EM power or disappears with PTA shuffling.

H2 Meta-Agent Efficacy $\langle \Phi \rangle$ (Xamã) $>$ (Sham) with $p < 0.01$ in simulation with plastic synapses. No statistical difference in final network synergy.

H3 Synthetic Coherence Neuromorphic chip's O-information becomes more negative during Sonens. Returns to baseline $< 24h$.

H4 Stellar Phase Transition Coherent fluctuations in P_S , P_H $\leq 3\%$ days after a gravastar formation. Within noise for $> 50\%$ of events.

H5 Axionic Resonance Neuromorphic axion detector's O-information correlates with P_S . Explained by thermal noise.

H6 Informational Gravity High-synergy system shows anomalous G_{eff} on a Cavendish balance. Null effect $> 5\sigma$.

H7 Informational Work (σ Measurement) A high-synergy group performs a complex task with greater metabolic efficiency, and σ converges to a stable value across systems. No efficiency difference; σ does not converge.

H28 Noosphere Hysteresis After removal of a Shaman attractor, a network with plastic synapses maintains an elevated synergy plateau, with decay time proportional to the learning rate. Immediate return to baseline.

H29 Meta-Holon Coupling Two coupled systems exhibit $\Phi_{\text{coupled}} > \Phi_1 + \Phi_2$ (synergy bonus). No integration gain.

H30 Directed Φ Evolution The rate of Φ change ($d\Phi/dt$) is proportional to the local O-information gradient (S). Φ evolution is a random walk, uncorrelated with S .

H31 Local Synergy Gradient The spatial gradient of O-information around a high- Φ subject is steeper and extends further than around a control. Gradients are identical.

H32 S as O-Information Gradient During a Sonen, the O-information gradient around the group is steeper and more extended than in a control condition. No difference.

H33 Synergy as Signature of Φ Systems with high Φ (IIT) exhibit consistently negative O-information; low Φ systems exhibit positive O-information. Weak or no correlation between Φ and O-information.

8. Conclusion

This revised framework addresses the structural criticisms raised by Sonnet 4.6. By integrating O-information as a local, measurable proxy for synergy, by reformulating the Field S as its gradient, by grounding σ in empirical measurement, and by adopting an adversarial experimental design, we have transformed the PIC from a speculative edifice into a rigorous, falsifiable research program. We invite the scientific community to prove us wrong.

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APPENDIX C: MULTI-AGENT SIMULATION CODE (UPDATED WITH PLASTIC SYNAPSES)

```
```python
import numpy as np

class HolonAgent:
 def __init__(self, phi_init, gamma=1.0):
 self.phi = phi_init
 self.gamma = gamma
 self.synapses = {}

 def forca_interna(self):
 return -self.gamma * self.phi * (self.phi**2 - 1)

 def atualizar_sinapses(self, agentes, alpha=0.1, dt=0.01):
 for id_pre, w in self.synapses.items():
 agente_pre = agentes[id_pre]
 if (self.phi > 0.5 and agente_pre.phi > 0.5) or (self.phi < 0.5 and agente_pre.phi < 0.5):
 w += alpha * (1 - w) * dt
 else:
 w -= alpha * w * dt
 self.synapses[id_pre] = np.clip(w, 0.0, 1.0)

class SimulacaoPentecostes:
 def __init__(self, N=100, alpha_social=0.1, alpha_stdp=0.1, sigma=0.05, dt=0.01):
 self.N = N
 self.dt = dt
 self.agentes = [HolonAgent(np.random.uniform(0.1, 0.5)) for _ in range(N)]
 for i, ag in enumerate(self.agentes):
 for j in range(N):
 if i != j:
 ag.synapses[j] = 0.1
 self.xama = HolonAgent(0.99, gamma=0.0)
 self.sham = HolonAgent(0.6, gamma=1.0)
 self.alpha_social = alpha_social
 self.alpha_stdp = alpha_stdp
 self.sigma = sigma
 self.historico_phi = []

 def passo(self, t, fonte_especial=None):
 agentes_ativos = self.agentes[:]
 if fonte_especial is not None:
```

```

 agentes_ativos.append(fonte_especial)
 for i, ag in enumerate(self.agentes):
 acoplamento = 0.0
 for id_pre, w in ag.synapses.items():
 agente_pre = agentes_ativos[id_pre] if id_pre < len(agentes_ativos) else
fonte_especial
 acoplamento += w * (agente_pre.phi - ag.phi)
 influencia_direta = 0.0
 if fonte_especial is not None:
 influencia_direta = 0.2 * (fonte_especial.phi - ag.phi)
 F_int = ag.forca_interna()
 ruido = self.sigma * np.random.randn()
 dphi = (acoplamento + influencia_direta + F_int) * self.dt + ruido * np.sqrt(self.dt)
 ag.phi = np.clip(ag.phi + dphi, 0.0, 1.0)
 for ag in self.agentes:
 ag.atualizar_sinapses(agentes_ativos, self.alpha_stdp, self.dt)
 self.historico_phi.append(np.mean([ag.phi for ag in self.agentes]))

def executar(self, rodadas=300):
 for t in range(rodadas):
 if 100 <= t < 200:
 self.passo(t, fonte_especial=self.xama)
 elif 50 <= t < 100:
 self.passo(t, fonte_especial=self.sham)
 else:
 self.passo(t, fonte_especial=None)
 return self.historico_phi
...

```

Histerese da Noosfera (H28): Após a remoção do Xamã, a rede mantém um patamar de coerência elevado devido às sinapses fortalecidas (LTP). O decaimento é função da taxa de aprendizado ( $\alpha_{stdp}$ ) e do ruído ( $\sigma$ ). A hipótese é falseada se a coerência retornar imediatamente à linha de base.

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#### APPENDIX D: NEUROMORPHIC HARDWARE CODE (UPDATED WITH O-INFORMATION)

```

``python
class MemristiveSynapse:
 def __init__(self, w_init=0.5, alpha=0.1, beta=0.05):
 self.w = w_init
 self.alpha = alpha
 self.beta = beta

```

```

def update(self, pre_spike, post_spike, dt=0.01):
 if pre_spike and post_spike:
 self.w += self.alpha * (1 - self.w)
 elif pre_spike and not post_spike:
 self.w -= self.beta * self.w
 self.w = np.clip(self.w, 0.0, 1.0)

class DetectorAxionNeuromorfico:
 def __init__(self, limiar_base=0.6):
 self.sinapse = MemristiveSynapse()
 self.limiar_adaptativo = limiar_base
 self.memoria_fundo = 0.0
 def processar_sinal(self, sinal_entrada, o_information_local):
 # O-information (negativa = sinergia) modula a taxa de esquecimento
 sinergia = -o_information_local # positivo se O<0
 taxa_esquecimento = 0.01 + (sinergia * 0.1)
 self.memoria_fundo = (1 - taxa_esquecimento) * self.memoria_fundo + taxa_esquecimento
 * sinal_entrada
 sinal_anomalo = max(0, sinal_entrada - self.memoria_fundo)
 pre_spike = (sinal_anomalo > self.limiar_adaptativo)
 post_spike = (o_information_local < -0.2) # Alta sinergia
 self.sinapse.update(pre_spike, post_spike)
 return pre_spike, self.sinapse.w
 ...

```

## APPENDIX H: INTERNAL ARCHITECTURE OF THE SHAMAN ( $\Phi$ MULTIPLICATION LAYERS)

### Layer Physical Substrate Function in PIC Measurable Proxy

1. Quantum Nuclear/electronic spins in microtubules (Orch-OR) Fundamental coherence bits  
Spin coherence time (if measurable)
2. Biochemical ATP cycles, DNA/RNA, protein synthesis Informational metabolism (energy  $\rightarrow$  order) Metabolic efficiency ( $\Delta U/\Delta \Phi$ )
3. Cellular Neurons, glia, organelles Local  $\Phi$  processing units Local O-information (multi-unit)
4. Energy Hubs Dantians / neurofascial plexuses Coherence amplifiers Bio-photon emission, electrical conductivity
5. Psychological Ego, shadow, self Double-well potential of psyche EFE scale, self-report
6. Integral (Shaman) Unified body-mind Complete holon irradiating synergy Global  $\Phi$  (IIT), O-information gradient (S)

H24 ( $\Phi$  Multiplication Architecture): The total  $\Phi$  of a Shaman is not the linear sum of the layers, but their integrated synergy. The "synergy bonus" ( $\$I_{\{ij\}}\$$ ) between layers is measurable via O-information. If  $\Phi_{\text{total}} = \sum \Phi_{\text{layers}}$  without bonus, the hypothesis is refuted.

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## APPENDIX L: THE COMPLETE SYNTROPY CIRCUIT

The journey of conscious information:

1. Descent ( $\eta \Phi_{\text{Source}}$ ): The Primary Source emits the Field S (O-information gradient).
2. Internalization ( $\sigma d\Phi$ ): The Shaman (complex holon) absorbs, processes, and amplifies the signal through its nested layers.
3. Horizontal Irradiation ( $\sum \alpha_{ij}$ ): The Shaman radiates coherence to the environment, elevating the synergy of nearby holons.
4. Ascent ( $\sum \gamma_i \Phi_i$ ): Information, enriched by experience, returns to the Source, closing the feedback loop.

Global Feedback Equation:

$$\frac{d\Phi_{\text{global}}}{dt} = \eta \Phi_{\text{Source}} - \mu \Phi_{\text{global}}^2 + \sum_i \gamma_i \Phi_i$$

This is the engine of Syntropy.

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